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PERIOPERATIVE MANAGEMENT

Perioperative Management of Patients with a Vagal Nerve Stimulator (VNS)

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Key Points

1. De-activate the Vagal Nerve Stimulator (VNS) ideally prior to **any** general anaesthetic, including short procedures. Neuroaxial block is not considered to be much safer.
    a. Primary perioperative risk is severe bradycardia or asystole due to unintended vagal stimulation.
    b. Secondary risk of vocal cord adduction & obstruction
2. If the VNS device remains active e.g. emergency surgery,
    a. Consider tracheal intubation as vagal nerve stimulation can cause unilateral vocal cord adduction and airway obstruction
    b. Have the device magnet immediately available or a pacemaker magnet – both temporarily deactivate it if necessary
    c. Continuous telemetry should be used.
3. Avoid monopolar diathermy near the device; use bipolar where possible.
4. Avoid cannulation of internal jugular or subclavian veins on the ipsilateral side to the device.
5. Ensure postoperative reactivation of VNS and document device status.

Introduction

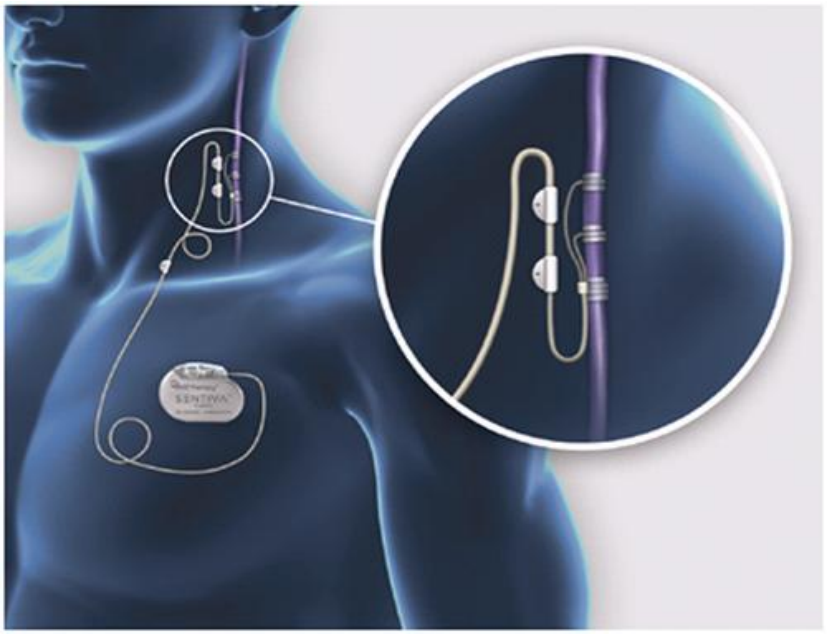


Figure 1 A Vagal Nerve Stimulator

A Vagal Nerve Stimulator (VNS) is an implanted neuromodulation device used primarily for epilepsy and treatment resistant depression. In the perioperative setting, VNS devices pose specific risks due to their parasympathetic effects (bradycardia, hypotension, syncope, unilateral vocal cord adduction) potential interference with anaesthetic equipment, and seizure management considerations.

Preoperative Assessment

Confirm VNS model, manufacturer (commonly LivaNova), indication, date of implantation, location of pulse generator and lead.

Review last device interrogation and baseline programmed settings (output current, duty cycle, frequency).

Ask the patient to bring their VNS magnet to theatre

Assess for VNS-related symptoms: cough, dysphonia, dyspnoea, obstructive sleep apnoea.

Continue antiepileptic medications perioperatively.

Evaluate cardiac history due to risk of bradyarrhythmia.

Discuss perioperative plan with neurology that put in the device. Confirm dates for deactivation prior to elective surgery.

Preoperative Deactivation of VNS

Deactivation of the VNS is recommended before any general anaesthetic, including short procedures. Because of case reports of severe bradycardia or asystole during induction or airway manipulation.

Methods:

Elective surgery = Formal device OFF via neurology/device technician. For YG patients this involves a trip to Walton.

Emergency Surgery = Temporary OFF by placing the magnet over the generator. If the VNS specific magnet is not available, a standard pacemaker magnet can be used. These are kept in recovery

In the emergency setting, as it is not always possible to arrange formal deactivation of the VNS device, it is reasonable to proceed with surgery with the device in the active mode. If left activated, the anaesthetist should be alert to the potential adverse side effects of vagal stimulation. If VNS-related complications arise, the device can be temporarily deactivated by securing the patient's personal VNS device magnet over the pulse generator location (commonly the anterior chest wall).



Figure 2 VNS Therapy System device magnet.

The magnet can be swiped across the generator by a patient or carer to deliver additional vagal stimulation if a seizure begins. Leaving the magnet in place over the generator results in deactivation.

Ensure the device remains OFF throughout the procedure and document status postop.

Intraoperative Management

Avoid monopolar diathermy near the generator; position return plate to direct current away from neck/chest.

Bipolar diathermy is preferable or avoided completely using other devices such harmonic scalpel, plasma wand.

Monitor for bradycardia; treat promptly with atropine or glycopyrrolate.

Consider endotracheal intubation due to VNS-related vocal cord adduction.

Maintain antiepileptic drug levels; have midazolam and levetiracetam (Keppra) available for seizure control.

Postoperative Care

Reactivate the VNS following surgery by removing the magnet from the pulse generator or device technician. For YG this involves a trip to Walton.

Confirm the patient has their VNS magnet prior to discharge.

Post deactivation VNS should be formally interrogated.

References

- Manufacturer clinical guidance (LivaNova).
- VNS Therapy™ is an epilepsy treatment designed to reduce the frequency, severity and recovery times of people suffering from seizures.
- Association of Anaesthetists – Perioperative management of patients with implanted neurostimulators.
- Management of vagus nerve stimulation therapy in the peri-operative period - Broderick - 2023 - Anaesthesia - Wiley Online Library
- Epilepsy Nurse Secretary – 0151 556 3537.

www.thewaltoncentre.nhs.uk/